

Hesham Salah

.NET Backend Developer

+20 112 510 0486 | hesham.salah.sa@gmail.com

[linkedin.com/in/theitshoo](https://www.linkedin.com/in/theitshoo) | github.com/theitshoo

Address: Cairo, Egypt

Professional Summary

Software Engineer with experience developing web applications and APIs, focusing on backend development and modern software engineering practices. Experienced with ASP.NET Core, Entity Framework Core, LINQ, and relational databases, along with additional knowledge of data access technologies and NoSQL databases. Familiar with DevOps practices and broader backend and frontend technologies.

Education

Bachelor of Computer Science

Sep 2022 – Jul 2026

Arab Open University - Egypt / Open University (UK)

- Completed an individual Graduation Project and achieved an **A** grade (**Excellent**).
- Graduation Project selected among the **Top 3 projects** of the year, showcased at the Graduation Projects Exhibition, and featured in the AOU Top Graduation Projects Gallery.

Projects

Wasiyyah: Islamic Will Management System 

Graduation Project

- Built a full-stack platform, that automate the end-to-end lifecycle from creation to executor-triggered activation for secure will Management.
- Designed a multi-role, state-driven system with controlled workflows and objection handling to ensure consistent and auditable will transitions.
- Developed a life-verification check-in engine using background jobs, automated reminders, and escalation logic for time-sensitive verification flows.
- Architected a modular .NET solution using layered dependency injection, EF Core code-first migrations, and SQL Server to ensure scalability and maintainability.
- Secured the platform with ASP.NET Core Identity (auth, RBAC, 2FA), CSRF protection, AES-encrypted file storage, and audit logging.

Itshoo Claim: Enterprise student deals API 

- Built a Clean Architecture REST API for student deal discovery and redemption with modular domains using ASP.NET Core Minimal APIs and Entity Framework Core with SQL Server.
- Implemented JWT authentication, ASP.NET Core Identity, refresh tokens, Google OAuth, role-based authorization, and FluentValidation.
- Developed deal discovery with search, efficient keyset pagination, and curated feeds for large deal listings and feed endpoints.
- Added production-ready API support: OpenAPI (Swagger/Scalar), Serilog logging, OpenTelemetry, health checks, rate limiting, and standardized error handling (RFC 7807).

Itshoo Reach: Simple Contact Manager 

Developed for Learning Purpose

- Designed an ASP.NET Core MVC solution with layered architecture (web, application, infrastructure) to isolate UI, business logic, and data access for easier testing and future changes.
- Shipped a full contact workflow (create/read/update/delete) backed by country-linked records to keep location data consistent and reduce duplicate or inconsistent entries.
- Integrated ASP.NET Core Identity for authentication and role-based authorization to protect contact data and workflows.

Skills

Technical Skills

- **Programming languages:** C#, Java, Python, Bash (Shell Scripting)
- **Backend Development:**
OOP, Data Structures & Algorithms, LINQ, ADO.NET, Dapper ORM, Entity Framework Core, SignalR, ASP.NET Core Web API, ASP.NET Core MVC, RESTful API, Swagger API, Dependency Injection(DI)
- **Security:**
ASP.NET Core Identity, Cookie-based Authentication, JWT (JSON Web Tokens), CSRF/Antiforgery, Role-Based Access Control (RBAC), Two-Factor Authentication (2FA), Serilog & Audit Logging
- **Database & Caching:** SQL Server, Relational Database Design, EF Core Migrations, MongoDB, Redis
- **Testing & DevOps:** Test-Driven Development (TDD), Unit & Integration Testing, Git/GitHub, Docker, Linux System Administration
- **Frontend Development:** HTML, CSS, Vanilla JavaScript, Razor Views (.cshtml)

Languages

- **Arabic**, Native Proficiency
- **English**, Professional Working Proficiency

Courses and Trainings

DevOps Engineering | DEPI Initiative

Mar 2024 – Dec 2024

- Mastered essential DevOps practices and technologies through an intensive program.
- Implemented real-world solutions by applying knowledge in hands-on projects.
- Developed technical expertise, enhanced freelancing capabilities, and strengthened soft skills with personalized coaching.

AWS Cloud Practitioner | NTI

Aug 2024 – Oct 2024

- Completed foundational training in AWS services, mastering cloud concepts and architecture best practices.
- Applied security measures and implemented solutions through hands-on labs for practical experience.

McKinsey Forward Program | McKinsey & Company

Nov 2023 – Apr 2024

- Developed leadership and digital skills, enhancing adaptability, communication, and problem-solving, through hands-on projects in a selective program.

Extra-Curricular Activities

Core Team Member | GDSC AOU

Sep 2023 – Aug 2024

- Collaborated globally with students, leaders, and experts to expand networking opportunities.
- Participated in hackathons and coding contests to strengthen technical skills, problem-solving, and teamwork.